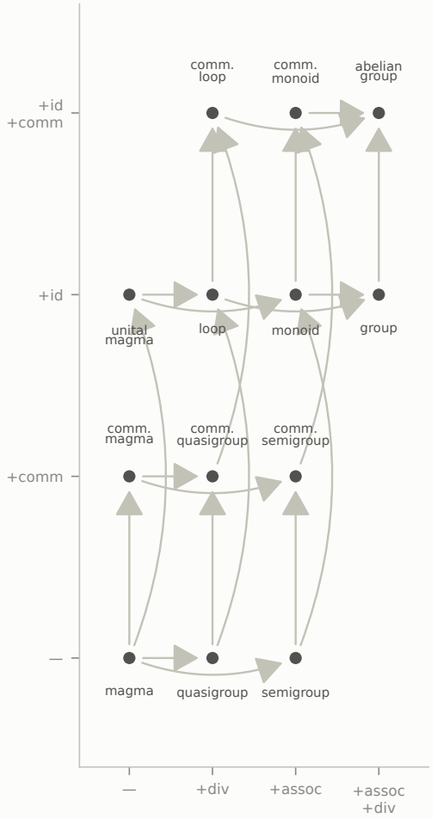
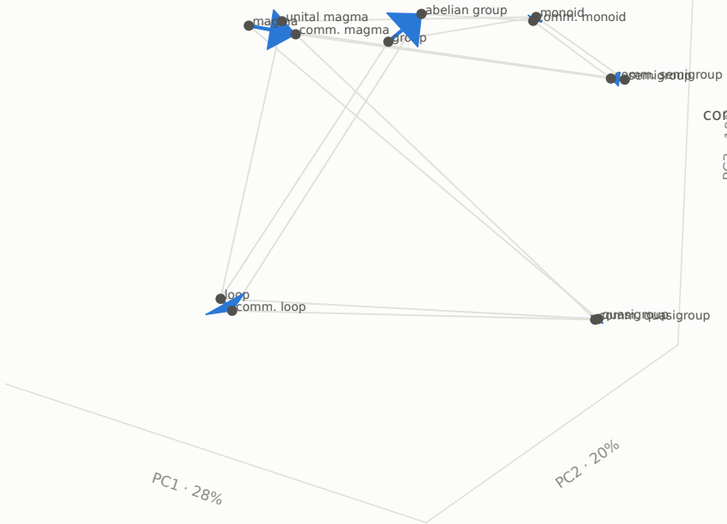
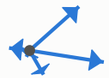
1 · Structures are sets of axioms



2 · Adding an axiom is an arrow + commutativity in Qwen2.5-1.5B (top-3 PCs)



3 · Line the arrows up



consistency 0.41; PC1–3 hold 66% of the variance

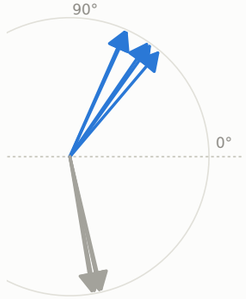
random-init twin

consistency 0.16; PC1–3 hold 42% of the variance

4 · Measure in the full space: angle of each arrow to the others' mean

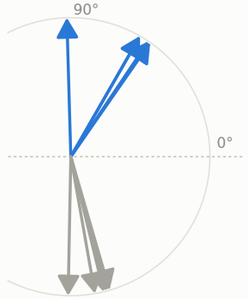
model above the dashed line (blue), twin below (grey); 0° = same direction, 90° = unrelated

+ associativity



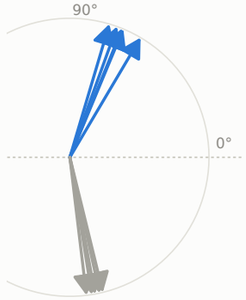
consistency 0.38 (p = 0.005)
twin 0.12

+ identity



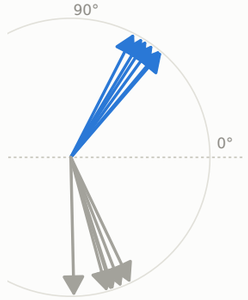
consistency 0.27 (p = 0.031)
twin 0.10

+ divisibility



consistency 0.25 (p = 0.041)
twin 0.10

+ commutativity



consistency 0.41 (p = 0.002)
twin 0.16